STAT210 Statistical Methods I: Linear Model

Lec 10: Multiple Linear Regression - Partial F-tests & Extra Sums of Squares

Zhaoxia Yu
Professor, Department of Statistics

2026-10-28

Section 1

Review

Review: Lecture 01-09

- Lectures 01-06: SLR — least squares estimate, inference (CI, prediction, t-test, F-test), diagnostics, transformations, ANOVA, lack-of-fit test.
- Lecture 07: MLR — least squares estimate.
- Lecture 08: MLR — inference (CI, t-test, linear functions, ANOVA).
- Lecture 09: sums of squares, more on F-tests and ANOVA.
- Lecture 10: partial (extra) sums of squares, partial F-tests, `anova()` model comparisons vs `aov()`, partial R^2.

Section 2

Partial F-tests (General)

Full vs Reduced models (nested)

We test whether a subset of k coefficients are zero given the remaining predictors are in the model:

$$H_0: \ \beta_{j_1} = \cdots = \beta_{j_k} = 0, \qquad H_a: \ \text{at least one of } \beta_{j_m} \neq 0.$$

Test statistic (SSE form)

Let SSE_F and SSE_R be the error sums of squares for the full and reduced models, respectively; let k be the number of tested coefficients, and df_F the residual df of the full model. Then

$$F = \frac{(\text{SSE}_R - \text{SSE}_F)/k}{\text{SSE}_F/df_F} \overset{H_0}{\sim} F_{k,\, df_F}.$$

Equivalent form (SSR)

$$F = \frac{(\text{SSR}_F - \text{SSR}_R)/k}{\text{SSE}_F/df_F}.$$

Section 3

Motivating Example: CDI

Model and data

Goal: predict unemployment percent (y) from x_1 = bachelor percent, x_2 = per capita income, x_3 = percent over 65.

```
cdi <- read.csv("data/CDI.csv")
lm.full <- lm(unemployment ~ bachelor + capita_income + X65plus, data = cdi)
summary(lm.full)
```

```
##
## Call:
## lm(formula = unemployment ~ bachelor + capita_income + X65plus,
##     data = cdi)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0438 -1.1638 -0.2989  0.6977 12.8957
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.214e+00  5.275e-01  17.469   <2e-16 ***
## bachelor      -1.802e-01  1.955e-02  -9.219   <2e-16 ***
## capita_income  5.040e-05  3.468e-05   1.453    0.147
## X65plus        2.021e-02  2.694e-02   0.750    0.453
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.964 on 436 degrees of freedom
## Multiple R-squared:  0.2991, Adjusted R-squared:  0.2943
## F-statistic: 62.03 on 3 and 436 DF,  p-value: < 2.2e-16
```

Partial F-test: income and X65plus given bachelor

Reduced model: $y = \beta_0 + \beta_1 x_1 + \varepsilon$.

```
lm.reduced <- lm(unemployment ~ bachelor, data = cdi)
anova(lm.reduced, lm.full)  # extra SS for (capita_income, X65plus) | bachelor
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ bachelor
## Model 2: unemployment ~ bachelor + capita_income + X65plus
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    438 1697.5
## 2    436 1681.8  2     15.71 2.0365 0.1317
```

Interpretation

The F-statistic is

$$F = \frac{(\text{SSE}_R - \text{SSE}_F)/2}{\text{SSE}_F/df_F}.$$

A small p-value indicates income and/or X65plus add explanatory power beyond bachelor.

Section 4

Extra Sums of Squares

Definition (subset given others)

$$\text{SSR}(\text{subset} \mid \text{others}) = \text{SSR}(\text{all}) - \text{SSR}(\text{others}) = \text{SSE}(\text{others}) - \text{SSE}(\text{all}).$$

CDI: compute components

```r
anova(lm.reduced)            # reduced model table
```

```
## Analysis of Variance Table
##
## Response: unemployment
##            Df  Sum Sq Mean Sq F value    Pr(>F)
## bachelor    1  702.05  702.05  181.15 < 2.2e-16 ***
## Residuals 438 1697.47    3.88
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(lm.full)               # full model table
```

```
## Analysis of Variance Table
##
## Response: unemployment
##                Df  Sum Sq Mean Sq  F value  Pr(>F)
## bachelor        1  702.05  702.05 182.0089 < 2e-16 ***
## capita_income   1   13.54   13.54   3.5099 0.06167 .
## X65plus         1    2.17    2.17   0.5631 0.45343
## Residuals     436 1681.76    3.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(lm.reduced, lm.full)   # extra SS and partial F
```

```
## Analysis of Variance Table
##
## Model 1: unemployment ~ bachelor
```

Section 5

Decomposition and Order (Type I SS)

Two predictors (x_1, x_2)

Decompose SST in two equivalent orders:

$$\begin{aligned} \text{SST} &= \text{SSR}(x_1) + \text{SSR}(x_2 \mid x_1) + \text{SSE}(x_1, x_2) \\ &= \text{SSR}(x_2) + \text{SSR}(x_1 \mid x_2) + \text{SSE}(x_1, x_2). \end{aligned}$$

Sequential (Type I) SS depend on order; the total $\text{SSR}(x_1, x_2)$ does not.

CDI: two-order comparison

```
model12 <- lm(unemployment ~ bachelor + capita_income, data = cdi)
model21 <- lm(unemployment ~ capita_income + bachelor, data = cdi)

anova(lm(unemployment ~ bachelor, data = cdi)); anova(model12)
```

```
## Analysis of Variance Table
##
## Response: unemployment
##            Df  Sum Sq Mean Sq F value    Pr(>F)
## bachelor    1  702.05  702.05  181.15 < 2.2e-16 ***
## Residuals 438 1697.47    3.88
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Variance Table
##
## Response: unemployment
##                Df  Sum Sq Mean Sq  F value  Pr(>F)
## bachelor        1  702.05  702.05 182.1910 < 2e-16 ***
## capita_income   1   13.54   13.54   3.5134 0.06154 .
## Residuals     437 1683.93    3.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(lm(unemployment ~ capita_income, data = cdi)); anova(model21)
```

```
## Analysis of Variance Table
##
## Response: unemployment
```

Takeaway

$$\text{SSR}(x_1) + \text{SSR}(x_2 \mid x_1) = \text{SSR}(x_2) + \text{SSR}(x_1 \mid x_2) = \text{SSR}(x_1, x_2).$$

For a partial F-test, enter the predictors being tested last.

Section 6

anova vs aov in R

Nested model comparison with anova()

```r
data(mtcars)
fit_wt    <- lm(mpg ~ wt, data = mtcars)
fit_wt_hp <- lm(mpg ~ wt + hp, data = mtcars)
anova(fit_wt, fit_wt_hp)   # extra SS for hp | wt (partial F)
```

```
## Analysis of Variance Table
##
## Model 1: mpg ~ wt
## Model 2: mpg ~ wt + hp
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1     30 278.32
## 2     29 195.05  1    83.274 12.381 0.001451 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

One-model ANOVA tables (aov() vs anova(lm()))

```r
aov(mpg ~ wt + hp, data = mtcars)            # Type I SS (sequential)
```

```
## Call:
##    aov(formula = mpg ~ wt + hp, data = mtcars)
##
## Terms:
##                        wt       hp Residuals
## Sum of Squares  847.7252  83.2742  195.0478
## Deg. of Freedom        1        1        29
##
## Residual standard error: 2.593412
## Estimated effects may be unbalanced
```

```r
anova(lm(mpg ~ wt + hp, data = mtcars))      # same sequential SS
```

```
## Analysis of Variance Table
##
## Response: mpg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## wt         1 847.73  847.73 126.041 4.488e-12 ***
## hp         1  83.27   83.27  12.381  0.001451 **
## Residuals 29 195.05    6.73
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Order matters for Type I SS

```r
aov(mpg ~ hp + wt, data = mtcars)
```

```
## Call:
##    aov(formula = mpg ~ hp + wt, data = mtcars)
##
## Terms:
##                        hp       wt Residuals
## Sum of Squares  678.3729 252.6266  195.0478
## Deg. of Freedom         1        1        29
##
## Residual standard error: 2.593412
## Estimated effects may be unbalanced
```

```r
anova(lm(mpg ~ hp + wt, data = mtcars))
```

```
## Analysis of Variance Table
##
## Response: mpg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## hp         1 678.37  678.37 100.862 5.987e-11 ***
## wt         1 252.63  252.63  37.561 1.120e-06 ***
## Residuals 29 195.05    6.73
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Notes

- `anova(fit)` with one model shows Type I (sequential) SS.
- `anova(reduced, full)` gives the partial F for nested models.
- `aov()` is tailored to classical ANOVA; extras include `Error()`, `TukeyHSD()`, `model.tables()`.

Section 7

Summary of Tests

Overall F-test in MLR

Hypotheses:

$$H_0 : \beta_1 = \cdots = \beta_p = 0,$$
$$H_a : \text{at least one } \beta_j \neq 0.$$

Statistic and null distribution:

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/p}{\text{SSE}/(n-p-1)} \overset{H_0}{\sim} F_{p,\, n-p-1}.$$

Single coefficient

$$t = \frac{\hat{\beta}_j}{\mathsf{SE}(\hat{\beta}_j)} \sim t_{n-p-1}, \qquad F = t^2 \sim F_{1,\, n-p-1}.$$

Partial F (subset size k)

$$F = \frac{(\text{SSE}_R - \text{SSE}_F)/k}{\text{SSE}_F/(n-p-1)} \overset{H_0}{\sim} F_{k,\, n-p-1}.$$

Section 8

Partial R^2

Definition (subset given others)

$$R^2_{\text{partial}} = \frac{\text{SSE}_R - \text{SSE}_F}{\text{SSE}_R} = \frac{\text{SSR(subset} \mid \text{others)}}{\text{SSE(reduced)}}.$$

Example: mtcars — hp given wt

```
SSE_wt    <- sum(residuals(fit_wt)^2)
SSE_wt_hp <- sum(residuals(fit_wt_hp)^2)
R2_hp_given_wt <- (SSE_wt - SSE_wt_hp) / SSE_wt
R2_hp_given_wt
```

```
## [1] 0.2992009
```

Example: mtcars — wt given hp

```
fit_hp <- lm(mpg ~ hp, data = mtcars)
SSE_hp <- sum(residuals(fit_hp)^2)
R2_wt_given_hp <- (SSE_hp - SSE_wt_hp) / SSE_hp
R2_wt_given_hp
```

```
## [1] 0.5643088
```

Takeaways

- Adding predictors cannot decrease overall R^2.
- Larger partial R^2 corresponds to larger partial F (more evidence against H_0).
- Use nested `anova()` for clear partial tests; Type I SS depend on order.

Section 9

Midterm Logistics

Midterm Logistics

- Format: in-class; closed book; two double-sided note sheets permitted.
- Coverage: through this lecture (SLR/MLR inference, ANOVA, overall & partial F-tests).
- Tips: interpret coefficients carefully; justify model comparisons; recall $F = t^2$ when the tested subset size is 1.